

Sol.

$$1. \quad B = \frac{\mu_0 I}{2r} = \frac{\mu_0 (e/T)}{2r}$$

$$\text{where } T = \frac{2\pi r}{v} \quad \text{but} \quad \frac{mv^2}{r} = \frac{e^2}{4\pi\epsilon_0 r^2}$$

$$\text{Therefore } B = \frac{\mu_0 e^2}{8\pi r^2 \sqrt{\pi m \epsilon_0} r}$$

2. Because of vertical magnetic field the charged particle moves in horizontal circle. And because of weight there is vertical shift in motion of the particle. hence the trajectory is a helical path.

3. B is zero everywhere inside the tube and maximum at the surface.

6. The expression of magnetic energy density is

$$\frac{B^2}{2\mu_0}.$$

The magnitude of magnetic field at end region is half that of at middle region. Hence the ratio of magnetic field energy densities at middle region to the end region is 4:1

7. The charged particle is moving in straight line at speed v w.r.t. the observer in backward direction.

The magnetic force on the charged particle is qBv ,

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10. $\uparrow B \Rightarrow \uparrow \tau \Rightarrow \uparrow$ current sensitivity

$$11. \quad \text{Current sensitivity} = \frac{NAB}{C}$$

12. magnetic field inside the core of toroid = $\mu_0 ni$

14. Gyromagnetic ratio = $\frac{\text{Magnetic moment}}{\text{Angular momentum}}$

$$= \frac{neh/4\pi m}{nh/2\pi} = \frac{e}{2m} = \text{constant.}$$

15. Let r be the radius of the circular loop

$$\therefore A = \pi r^2 \text{ or } r = \sqrt{\frac{A}{\pi}}$$

Magnetic field at the centre of the loop is

$$B = \frac{\mu_0 I}{2r} = \frac{\mu_0 I}{2\sqrt{\frac{A}{\pi}}} \text{ or } I = \frac{2B}{\mu_0} \sqrt{\frac{A}{\pi}}$$

So magnetic moment

$$M = IA = \frac{2B}{\mu_0} \sqrt{\frac{A}{\pi}} \times A = \frac{2BA\sqrt{A}}{\mu_0 \sqrt{\pi}}$$

16. NCERT P.No. - 135 / Table 4.1

17. Electrostatics field lines never form close loop

18. NCERT P.N. - 153 / Magnetic confinement

$$20. \quad \frac{E}{B} = c \quad \& \quad \sqrt{\mu_0 \epsilon_0} = \frac{1}{c}$$

21. Net force on current carrying loop placed in uniform magnetic field is always zero.